Apartment 50: the receipt on the table

Fictional examination staging inspired by Landau's theoretical-minimum tradition and Bulgakov's characters. These are invented lines, not historical quotations.

Behemoth pushed a receipt across the table. Two zeros gleamed on it.

"Perfect," he said. "I have already printed the admission ticket."

Landau covered the zeros with his hand. "Show me the carrier."

Woland covered the signature box. "Show me who checked it."

Behemoth turned the receipt over. The reverse said `unsigned_lab`. Below it were three unchecked conformance laws and an admission verdict: `Refused`.

"The arithmetic may be excellent," said Landau. "Now explain what the result actually permits you to claim."

The examination begins with small distinctions: bytes and meanings, graph edges and declared faces, execution and admission. Then you must reconstruct the four-vertex calculation, defend it with an independent lower bound, and stop a plausible forgery. The final round asks you to design the evidence another person would need to check your work.

Read the student guide, attempt the quiz, and open the instructor key only after an attempt. Hints are allowed; invented evidence is not. A clearly explained missing obligation earns more credit than a confident unsupported success label.

Behemoth may help you find near misses. Keep him away from the release button.

This companion teaches the finite contracts in Paper 004. Its examples grant no operational authorization and assert no physical or model-internal result.

Student guide: exact calculations and the custody of evidence

Learning companion for Paper 004, v0.1.0. Read this before the quiz. Solutions are in the separate instructor key. You need basic vectors, sums, absolute values, and the idea of a cryptographic digest. No Rust or Lean installation is needed for the hand work.

By the end, you should be able to calculate the paper's finite fixture, distinguish a mathematical witness from a hash commitment, predict several typed refusals, and state what evidence a reproducible native run would require.

1. Five questions that belong to different checks

Question	Relevant evidence	What remains separate
Are these the committed bytes?	Named encoding profile and comparison with a retained digest anchor	Whether the bytes state a true claim
Is this the declared finite problem?	Vertices, oriented edges, faces, coefficient domain, weights and manifest anchor	Whether it models an external system faithfully
Does this candidate equal the replayed normal form?	Exact solve, replay and candidate comparison	Whether the producer is authentic
Did the intended executable produce this result?	Independently trusted binary selection and execution/provenance evidence	Whether source-to-binary construction is reproducible
May this result be admitted?	The action's conformance and attestation policy	Merely obtaining zero numerical defects

Avoid carrying an answer from one row into another. A hash can bind a false statement. A correct calculation can have refused admission. An unsigned outcome can be internally consistent without establishing its producer.

2. Reconstruct the carrier before calculating

Use vertices `0,1,2,3`, oriented edges in the order

`(0,1), (1,2), (2,0), (1,3), (3,0)`

and faces `(0,1,2)` and `(0,1,3)`. Unit edge penalties are used. For a vertex potential `f`, the edge difference on `(u,v)` is `f(v)-f(u)`.

The before cochain is `c=(3,2,2,1,1)`. In this fixture:

$$\Phi_1(c) = \min_f \sum_e |c_e - (d_0 f)_e|, \qquad \Gamma_2(c) = |c_{01} + c_{12} + c_{20}| + |c_{01} + c_{13} + c_{30}|.$$

The faces are data, not decorations on the graph. A graph with the same edges but one fewer face changes `d_1` and therefore changes what the face check observes. Its carrier identity must change.

3. A worked upper bound and a matching lower bound

Try `f=(0,-2,-2,-1)`. The calculation is:

Edge	Before `c`	Difference `d f`	Residual `c-d f`
01	3	-2	5
12	2	0	2
20	2	2	0
13	1	1	0
30	1	1	0

The residual cost is `7`. This establishes `Phi1 7`; trying one potential does not yet prove optimality.

For the lower bound, assign `y=(1,1,1,0,0)` to the oriented edges. It circulates around `0→1→2→0`: inflow equals outflow at each vertex. Thus `y·d g=0` for every potential `g`. Each coordinate has absolute value at most its unit penalty, so

$$\sum_e |c_e - (d_0 g)_e| \geq y \cdot (c - d_0 g) = y \cdot c = 7.$$

Upper and lower bounds meet. **Phi1=7 exactly.** No timing result, approximate optimizer, or native receipt is needed to verify this small mathematical fact.

The two face sums are 7 and 5, hence Gamma2=12. The graph has `5-4+1=2` independent cycles, and these two faces have independent boundaries spanning them. Under the paper's declared solve, the final normal form is zero; its final Phi1 and Gamma2 are both zero. This does not make the original vector `c` a gradient. It distinguishes the before calculation from the normal form returned by the solve.

4. Exact execution can end in refused admission

For the canonical host fixture, the expected execution label is `EXECUTED_REPLAYED_EXACT_LAB_EVIDENCE`. The candidate matches the replayed normal form, and the process consumes two internal permits, for the letters `E,T`.

The admission result is still `Refused`: conformance is `Partial`, three laws are unchecked, and attestation is `unsigned_lab`. The codes are `PRIME000_CONFORMANCE_PARTIAL` and `UNSIGNED_LAB_ATTESTATION`.

A good display keeps execution, numerical summaries, conformance, attestation and admission together. Printing "verified: zero" loses information the consumer needs.

5. Canonical bytes need a named contract

The paper names three layers:

Layer	Rules to remember
Domain canonical algebra	Tagged types: integer `7` and string `"7"` remain distinct
`PacketCodec` helper	NFC normalization, then tagged domain encoding; its `verify` checks payload digest and authority, not every sealed envelope field
Restricted host's outer JSON	`omega-sorted-json-no-float-v1`; exact arithmetic coefficients use minimal i128 strings

RFC 8785 has its own primitive serialization and sorting rules and preserves Unicode string data as supplied. Consider U+00E9 (`é`) and U+0065 U+0301 (`e` followed by a combining acute accent). They look alike but their direct UTF-8 bytes differ. NFC maps both to U+00E9. A normalization policy is therefore part of the meaning of a digest comparison.

The type of a mathematical quantity and its permitted wire spelling are separate. The restricted host computes internally with `BigRational`, but its before/candidate wire fields accept integer strings. Internal rational arithmetic does not make `"1/2"` a permitted spelling in those fields. The broader dispatcher has distinct action contracts.

6. Three ways a plausible receipt can mislead

A changed carrier with a stale anchor. Recomputing an outer request hash does not repair a mismatch between a derived manifest and its independently bound expected carrier hash.

A changed candidate with matching surrounding hashes. The host still compares its values to the replayed normal form. A candidate digest cannot replace that comparison.

A changed outcome with a recomputed root. The unkeyed root can become consistent with the edited bytes. That alone supplies neither a fresh native replay nor producer authentication. If an attacker can also replace the expected anchor, a comparison against that replacement provides no independent protection.

Write down which party controls each value and which anchor the checker actually trusts. "It has a hash" is not a threat model.

7. Interface and resource boundaries

The default host, runtime dispatcher and foundation library are distinct. The environment override `OMEGA_SYMBOL_GAME_HOST` belongs to the host seam. The graph tutorial adapter instead needs `OMEGA_FUNCTIONAL_GRAPH_BIN` and `OMEGA_FUNCTIONAL_GRAPH_SHA256`. Advertising an action does not configure another executable.

The restricted host has a 1,048,576-byte request ceiling, dimension limits of 4,096, and four dense-matrix cell checks at 1,048,576 cells each. The dispatcher uses a separate 65,536-byte input ceiling. Its bounded reader takes at most its limit plus one witness byte.

The declared timeout is validated metadata, not an in-process watchdog. A process wrapper must separately enforce and record any required wall-clock termination. Finite dimensions do not bound the bit complexity of all exact intermediate arithmetic.

In the package governor, an unset graph BIN chooses exact Python enumeration. A configured native call that refuses or fails must propagate that outcome. A native-required engine action cannot be relabeled as successful native execution by replacing it with an offline Python answer.

8. Read test receipts as bounded evidence

Retain the command, date, exit status, source/input hashes, interpreter/toolchain, native selection and binary digest when used, and passed/failed/skipped counts. A scripted executable checks adapter behavior; it does not independently validate native arithmetic. Two wrappers calling one solver are not two independent solvers.

The package's Lean modules concern specific finite mathematical statements. They do not prove the host protocol or connect every transport field to the implementation. An axiom census records named declarations and their axiom use, not the truth of every sentence in a paper.

Now attempt the quiz. Use the hints after an honest first attempt. For each answer, name the carrier, calculation/check, and limit of the resulting claim.

Instructor key: Paper 004 theoretical minimum

Instructor key for v0.1.0. Keep this separate from the student guide. These answers concern the declared finite problems and named interfaces. They do not supply a native execution receipt or learner-study result.

Round 1

1. `ab` and `ba` have different byte sequences; length carries no digest-equality guarantee. Under the encoding and cryptographic assumptions, matching an independently retained expected digest supports an integrity comparison with the committed representation. It does not establish mathematical truth, producer identity, source-to-binary provenance, freshness, or operational admission. Digest equality is not an unconditional mathematical proof that two arbitrary byte strings are identical: collision assumptions matter.
2. Execution: `EXECUTED_REPLAYED_EXACT_LAB_EVIDENCE`. Admission: `Refused`. Codes: `PRIME000_CONFORMANCE_PARTIAL`, `UNSIGNED_LAB_ATTESTATION`. The exact zero summaries and replayed candidate relation concern the finite calculation. Admission concerns additional policy/evidence obligations; it remains refused. Three conformance laws are unchecked and attestation is `unsigned_lab`.
3. Tagged encoding preserves type information, so integer and string images differ. A parser or consumer may apply different arithmetic, validation or comparison rules to the two values. Erasing their distinction can bind ambiguous data to one apparent identity.
4. The face set, face operator `d1`, closure problem and carrier manifest change. For this before vector, the remaining face sum is 7 rather than the pair of sums 7 and 5. Recomputing the outer request ID cannot repair a mismatch with the old expected carrier manifest/hash. The restricted host must refuse `CARRIER_MANIFEST_MISMATCH` after rebuilding the declared carrier's manifest.
5. No. `OMEGA_SYMBOL_GAME_HOST` selects through the host seam; the graph tutorial needs `OMEGA_FUNCTIONAL_GRAPH_BIN` and `OMEGA_FUNCTIONAL_GRAPH_SHA256`. An adapter name is discovery information. It is not configuration of the other binary, a digest comparison, an executed calculation, or a verified result.
6. Warranted: named local Lean declarations and their reported axiom dependencies concern the source statements in those modules, subject to successful replay with the pinned toolchain. Unwarranted: the census proves the native host's correctness; or it establishes an end-to-end correspondence between every Python object, Rust action and exported receipt field. A topology lemma is not a test of transport serialization.

Round 2

7. `d0f=(-2,0,2,1,1)`; residual `(5,2,0,0,0)`; unit-weight L1 cost 7. This is a feasible potential, so `Phi1 7`. Optimality needs a lower bound valid for every potential.
8. Use `y=(1,1,1,0,0)`. At vertex 0, outgoing 01 and incoming 20 cancel; at 1, incoming 01 and outgoing 12 cancel; at 2, incoming 12 and outgoing 20 cancel; vertex 3 has zero flow. Thus `d0 y=0`; every `|y_e| 1`; `y·c=7`. For any potential `g`, `Σ|c-d0g| y·(c-d0g)=7`. Together with question 7, `Phi1=7`. Check orientation signs; changing one edge's direction requires changing the corresponding circulation coordinate.
9. Face 012: `3+2+2=7`. Face 013: `3+1+1=5`. Gamma2=`|7|+|5|=12`. The face-boundary vectors `(1,1,1,0,0)` and `(1,0,0,1,1)` are independent. The connected graph's cycle-space dimension is `E-V+1=2`, so they span it. Under the declared solve, no first-cohomology direction remains in the final normal form. The original `c` is not a gradient: gradients telescope to zero on each cycle, while `c` has nonzero cycle sums. Do not confuse the solver's final normal form with an arbitrary gauge change of the original data.
10. The cycle sum is 3. For penalties `(1,2,2)`, cost is at least `min(a)·|3|=3`; residual `(3,0,0)` attains it. Projected cochain `c-r=(-2,1,1)` sums to zero. An attaining potential in vertex order is `(0,-2,-1)`. The cheapest edge is unique, so the residual/projected cochain is unique; potentials differ by an additive constant. With unit penalties, residuals `(3,0,0)` and `(0,3,0)` are distinct, both have sum 3 and cost 3. Uniqueness fails.
11. Composed UTF-8 is `c3 a9`; decomposed UTF-8 is `65 cc 81`. NFC sends both to the composed string. RFC 8785 preserves supplied Unicode string data; its primitive/sorting rules also differ from the tagged helper. The function name and visually identical output establish no standards-conformance claim.
12. Recomputing the root demonstrates consistency between an edited record and that newly supplied unkeyed root. It does not demonstrate a fresh replay or authenticated producer. Nor does it fulfill admission requirements. Comparison against a separately retained original root can detect an edit; replacing that anchor defeats this particular comparison. The host's internal solve/replay evidence is a separate object and event.
13. Products: `d0=2048×512=1,048,576`; `d1=1×2048=2,048`; augmented=`2048×513=1,050,624`; closed-system=`1×513=513`. The augmented matrix exceeds the cell ceiling, so the carrier fails the preallocation fence despite satisfying every individual dimension limit. The host returns `RESOURCE_FENCE_EXCEEDED` before allocating the fenced matrices. Timeout metadata alone cannot kill a process; enforce a deadline in the process owner and record a timeout as such.
14. `"0"` and `"-7"` are minimal signed i128 strings and are permitted. `"+7"` and `"007"` are nonminimal encodings of 7; `"-0"` is a nonminimal encoding of 0; `"1/2"` is not an integer string. Their inclusion in

the before vector causes `BEFORE_RESIDUAL_INVALID`; in the candidate vector, `CANDIDATE_RESIDUAL_INVALID`, assuming earlier checks pass. `BigRational` internally does not widen these restricted wire fields. Any purported dispatcher alternative must name its own action/input contract.

Final challenge rubrics

15. One valid construction copies an outcome, changes a claimed exact final value, and derives a new outcome root using the protocol's exact framing/canonicalization. A root-only checker can accept the new root without rerunning the computation. The attacker need not invert a digest; it computes a commitment to its own forged bytes. If it can replace the expected anchor too, matching those two attacker-controlled values adds no independent assurance.

    To resist this, a consumer must preserve independent anchors and obtain trustworthy producer/execution evidence or rerun the selected checker on the declared request. A pinned executable hash establishes selected bytes under the execution assumptions; source provenance and reproducible build require more. A challenge alone is not authentication, and an unsigned record that echoes a nonce does not prove who executed anything. Authentic replay still leaves the fixture's three unchecked conformance laws and unsigned attestation, so admission remains refused. Full credit requires an actual false field or event claim, the check that would still pass, and the unmet obligations; "hashes are weak" is insufficient.

16. The partial CUT is refused as `IncompleteProposal`, with state unchanged. Exactly one exact value for every declared edge is required. A complete all-zero proposal can be scored on the declared carrier; its cycle minimum is zero. It changes the problem's claims. Whether that edit is a permissible representation repair or a false rewriting of observations requires the task's independent semantics/policy. Zero cost in the altered instance supplies no evidence that immutable external data was incorrect. Do not interpret an omitted edge as zero.

17. For rational directed cycles, use the independently derived identity `min_f Σa_i|c_i−d0f_i|=min(a_i)|Σc_i|`. Check the solver's scalar against that value and check its returned potential's cost; include negative/zero sums, unique/tied minimum penalties, exact fractional data, and reversed orientations handled consistently. A tie permits different valid potentials; equality of full output vectors is not the correct universal criterion.

    On another graph, retain a feasible potential `f` and dual vector `y`. Independently check `B y=0`, `|y_e| a_e`, residual `r=c−Bf`, and exact equality `Σa_e|r_e|=y·c`. These checks certify optimality for that declared graph because the primal and dual bounds meet. The four-vertex graph in questions 7–8 is one such example. If bounds do not meet, the certificate

has not established optimality. Certificate checks do not prove universal solver correctness, physical fidelity, producer authenticity or source-to-binary custody. Full credit requires distinct computation/check paths rather than calling the same routine twice.

18. A strong evidence plan includes the following:

Item	Required content
Obtainable source	Immutable public source revision, relevant workspace/supporting crates, Cargo lockfile, license and instructions
Declared fixture	Ordered vertices/edges/faces, unit penalties, before/candidate vectors, coefficient domain and budgets
Trusted anchors	Expected carrier manifest, registry/source bindings and packet/request identities; provenance of independently retained anchors
Encoding	Exact named profile, tagged payload rules, minimal i128 inputs, hash framing and field exclusions
Executable and execution	Toolchain/platform, actual binary selection and SHA-256, environment override, command, exits and launcher-enforced wall-clock limit
Results	Exact summaries 7/12 then 0/0, candidate match, execution status, partial laws, E/T writes, unsigned attestation and refused admission codes
Receipt custody	Input/source/log hashes, commands and timestamps, separate passed/failed/skipped counts, and a clear declaration of checks not executed

Suitable hostile controls include:

Mutation	Expected observation
Delete a face, retain old carrier anchor, recompute request ID	`CARRIER_MANIFEST_MISMATCH`

Mutation	Expected observation
Change candidate to another vector, recompute outer identity	`CANDIDATE_NOT_REPLAYED_NORMAL_FORM`, if earlier checks pass
Alter a returned field without replacing its root	Root check fails
Alter a returned field and recompute unkeyed root	Root consistency can pass; fresh replay/authentication remain unestablished
Duplicate a JSON key	Intake refusal under duplicate-key rejection
Add an unknown request field	Intake refusal under the declared schema
Add one byte beyond the interface's request ceiling	Interface-specific intake refusal; preserve which ceiling was tested
Use nonminimal coefficient `"007"` with all earlier identities updated	Exact input parse refusal
Replace a configured graph executable with a digest-mismatched file	Graph tutorial adapter refuses; no numerical fallback

Require at least five controls with predicted checks and ordered preconditions. A test with a stale outer request ID that fails early does not demonstrate the deeper carrier or candidate check. Scripted hosts are labeled as adapter tests. Historical September results retain their dates; a fresh run records its actual revision and bytes; an unavailable executable is not counted as a pass. A private GitHub path alone does not meet this question's public-deposit requirement.

Marking advice

Reward exact witnesses, declared assumptions and precise missing obligations. Deduct for promoting source expectations into execution, treating a recomputed root as a signature, presenting a Python fallback as native evidence, or using a changed carrier without disclosing the change. Accept different valid primal/dual witnesses and sensible alternative custody plans when their domains and checks are explicit.

Landau's theoretical minimum: Paper 004

Companion for v0.1.0. Read the student guide, solve these questions, then consult the separate instructor key. All Apartment 50 dialogue is fictional.

Questions 1–6 establish vocabulary; 7–14 require calculations or precise contract decisions; 15–18 are the final challenge round. Work over exact integers/rationals unless a question explicitly names bytes. A correct explanation of unavailable evidence is a valid answer; silently replacing the problem is not.

Round 1: the receipt has arrived

1. Behemoth compares `ab` and `ba` and announces that equal lengths mean equal digests. Identify the mistake. If two artifacts match a retained expected digest, what does that establish, and what does it leave open?
2. A proposal has zero final Phi1 and Gamma2, partial conformance, and unsigned laboratory attestation. Give its execution label, admission verdict and two admission codes. Which part is a mathematical result and which part is a policy decision?
3. An integer `7` and string `"7"` pass through the domain canonical algebra. Why must their encodings remain distinguishable? Explain one problem caused by erasing this distinction.
4. Landau removes face `(0,1,3)` while keeping every graph edge. What changes mathematically? A caller recomputes the outer request hash but retains the old carrier anchor: which check must reject it?
5. The runtime description advertises an adapter. Does this configure the graph tutorial executable? Name the two environment-variable families and the evidence missing from the advertisement.
6. Sixty topology lemmas and a clean axiom report are present. State one warranted claim and two unwarranted claims about the native host.

Round 2: Landau wants the numbers

Use the guide's four-vertex carrier and edge order for questions 7–9.

7. Starting with `c=(3,2,2,1,1)` and `f=(0,-2,-2,-1)`, calculate all five edge differences, all residuals, and the residual cost. Explain why this only gives an upper bound until another argument is supplied.
8. Construct a dual circulation that proves a lower bound of 7. Check balance at every vertex, the unit coordinate caps, and its pairing with `c`. Conclude the exact minimum.
9. Calculate both face sums and Gamma2. Explain why spanning both graph cycles with declared faces supports the zero final normal form under the declared solve without implying that the before cochain was already exact.
10. For the cycle claims `(1,1,1)` and penalties `(1,2,2)`, derive the minimum and a minimizing projected cochain. Now change the penalties to `(1,1,1)`

and construct two distinct optimal residuals. What happens to uniqueness?

11. Compare UTF-8 bytes for U+00E9 and U+0065 U+0301. What does NFC do? Explain why the helper's NFC step does not establish RFC 8785 conformance.
12. An attacker edits a returned outcome and recomputes its unkeyed root. Distinguish an internally consistent record, a fresh replay, an authenticated producer and an admitted result. Which of these has the attacker shown?
13. Consider a declared carrier with 512 vertices, 2,048 edges and one face. All dimension limits are satisfied. Evaluate the four cell products for `d0`, `d1`, the augmented matrix, and the closed-system matrix. Is the preallocation fence satisfied? Also explain why a valid timeout field does not terminate a long solve.
14. Which restricted before/candidate wire spellings are permitted: `"0"`, `"-7"`, `"+7"`, `"007"`, `"-0"`, `"1/2"`? Give a reason for every refusal. Do not infer the restricted wire contract from the internal arithmetic type.

Final round: Woland checks the missing obligations

15. **The perfectly rehashed impostor.** You control every field in an exported JSON outcome and can compute every unkeyed digest. Construct a plausible forgery that passes a root-consistency check but does not demonstrate native replay. Identify the independent evidence a consumer would need to defeat your attack. Finally explain why even a genuine replay of this fixture still has refused admission.
16. **The changed challenge.** Begin with cycle claims `(1,1,1)`. A relational CUT proposes zero on only two edges and omits the third. Describe the correct outcome and state behavior. Compare it with a complete all-zero proposal. Explain why accepting the latter cannot establish that an immutable external observation was wrong.
17. **Two wrappers, one oracle.** Two Python wrappers agree on 100 fixtures but call the same calculator. Design a genuinely independent check for the cycle family and one exact primal/dual check for a graph outside that family. State what the checks establish, which input cases they cover, and what source/binary/provenance obligations remain.
18. **The public reproducibility challenge.** Build an evidence-custody plan for another student to reproduce the native four-vertex path from a standalone public deposit. Specify source/dependencies, fixture and anchors, wire profile, launcher/binary identity, toolchain, commands, process limits, results and refusal fields. Define at least five hostile controls, each with a predicted check or refusal. Keep historical results, your fresh run, skipped tests and unexecuted dependencies distinct.

Optional hints

Question	First hint
7	On edge 20, subtract `f(2)` from `f(0)`
8	A unit circulation around 012 has no net vertex divergence
9	The two faces share edge 01 but have different remaining edges
10	Every residual has the same cycle sum; put all mass on a cheapest edge
11	The composed character uses two UTF-8 bytes; the decomposed pair uses three
12, 15	Ask who supplied the expected anchor and who executed the checker
13	Multiply the edge count by vertices plus faces
14	Parsing must return the same spelling when the parsed integer is serialized minimally
16	An omitted value is not an explicit zero
17	An upper bound plus an independently checked matching lower bound is a certificate
18	Include one altered face, one altered candidate and one oversized request

Grading

Each of questions 1–14 is worth 4 points: 2 for the correct result, 1 for the check or derivation, and 1 for the evidence limit. Each final challenge is worth 11 points: 4 for the concrete construction, 4 for checks and expected outcomes, and 3 for custody and scope. Total: 100.

Missing proof or execution is an open obligation, not a fabricated success. Full credit requires preserving the declared problem and recognizing when a different interface, input, or calculator would change it. The marking scheme is instructional; it is not a validated learner assessment.